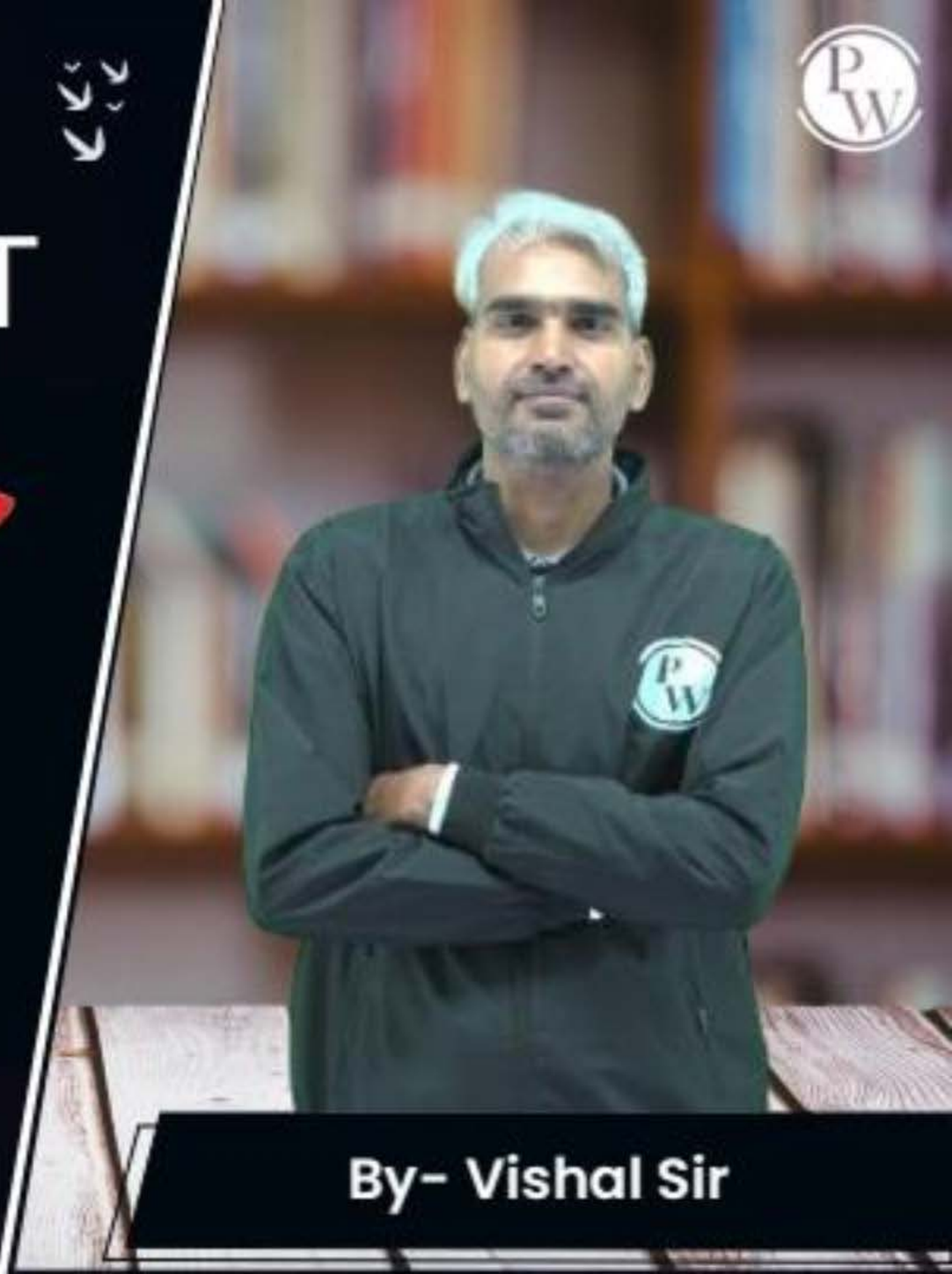
Computer Science & IT



Database Management System

ER Model

Lecture No. 01



By- Vishal Sir

Recap of Previous Lecture



Topic — B+ Tree insertion

Topic — B+ Tree deletion

Slide

Topics to be Covered



- **Topic** ER model & ER diagram
- **Topic** Terminologies in ER diagram

Slide



Topic : Entity-Relationship Model (ER-Model)

- ER Model is a high-level data model diagram which defines the conceptual view of the database (OR) that gives the graphical representation of the logical structure of the database.

- Entity-Relationship model is based on objects called entities, and relationship among these entities.



Topic : Entity

- An entity refers to any object having either a physical existence or a conceptual existence. For example, a school, a person, a house, a university, a company or a job.



Topic : Entity Set

An entity set is a collection or set of all entities of a particular entity type at any point in time.

"Student" Entity set

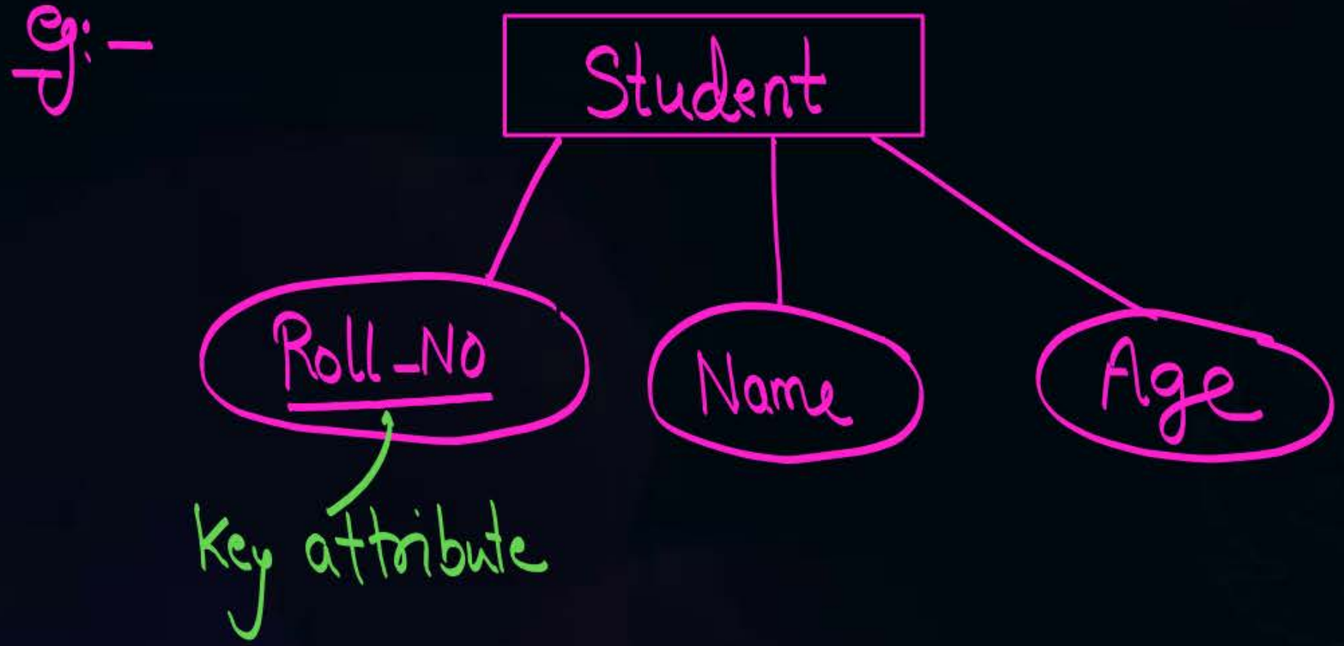
Roll_no	Name	Age
1	Ram	20
2	Shyam	19
3	Mohan	20
4	Arjun	19

This complete table is referred as "Student Entity Set" and each row represents an "entity".



Topic : Entity Set

An entity set is represented by a rectangle in ER diagram

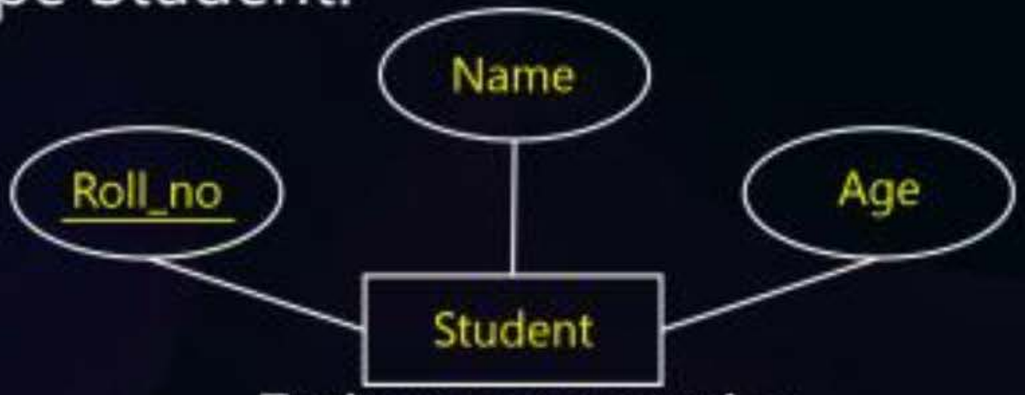




Topic : Attribute

- Each entity is identified uniquely by the values of its attributes.
- An attribute in an Entity-Relationship Model describes the properties or characteristics of an entity. It is represented by an oval or ellipse shape in the ER diagram. Every oval shape represents one attribute and is directly connected to its entity set which is rectangle in shape.

For example, roll_no, name, and age are the attributes which define entity type Student.



Entity representation

Roll_no	Name	Age
1	Ram	20
2	Shyam	19
3	Mohan	20
4	Arjun	19



Topic : Types of Attributes

i. Simple attribute

ii. Composite attribute

iii. Single-valued attribute

iv. Multi-valued attribute

v. Key attribute

vi. Derived attribute

vii. Partial key attribute



Topic : Types of Attributes

- (i) **Simple Attributes:** An attribute which contains an atomic value and cannot be divided further is called a simple attribute. Example, Roll_no, Sex, etc.
- (ii) **Composite attribute:** An attribute which can divided further. It is an attribute which is composed of several components. Example name, address, etc. Name ≡ first_name, middle_name, last_name
- (iii) **Single-valued attribute:** An attribute which can have only one value.
- (iv) **Multi-valued attribute :** An attribute which has multiple values (more than one value)
- (v) **Key attribute :** An attribute which can identify all entities uniquely.
- (vi) **Derived attribute :** An attribute whose value can be derived or found from any other attribute is known as derived attributes. { if D.O.B is stored attribute, then Age may be derived attribute }
- (vii) **Partial key attribute :** An attribute which can identify a group of tuples.



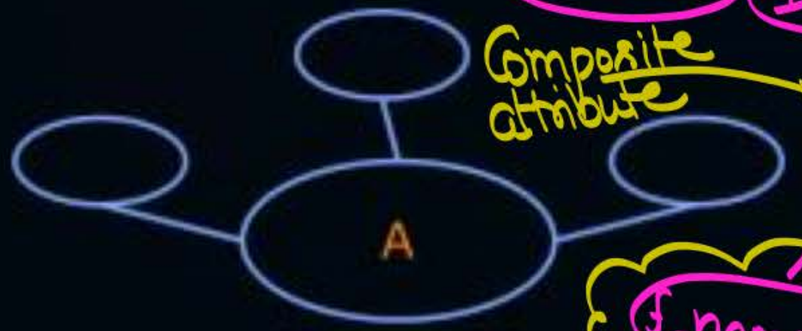
Topic : Types of Attributes



Attribute
(Simple + Single valued)



Multivalued Attribute



Composite Attribute



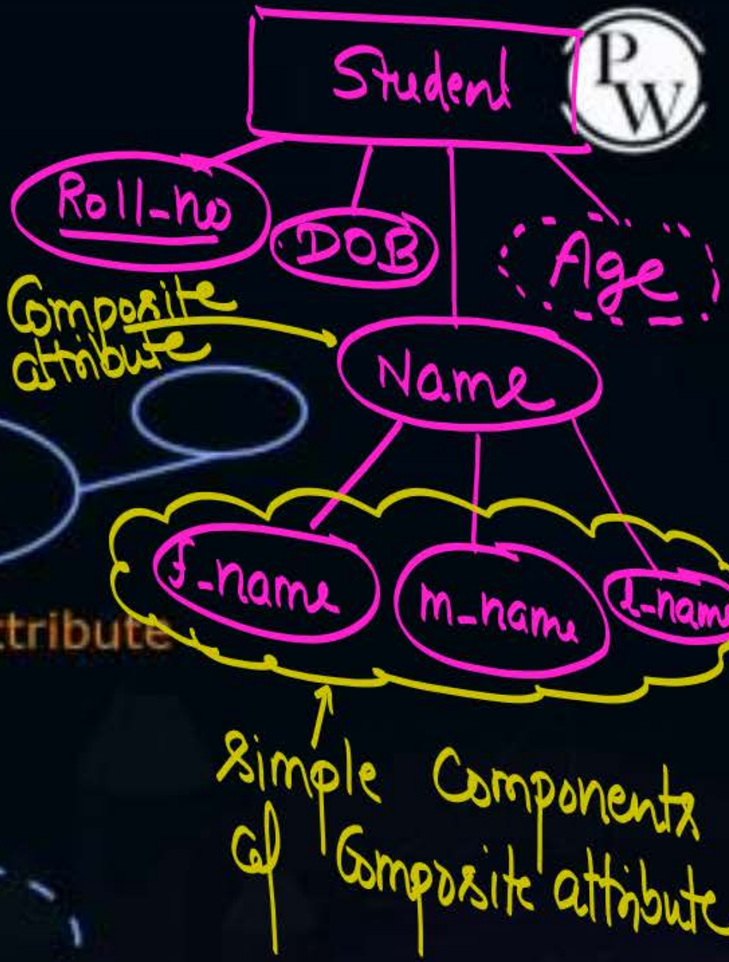
Key Attribute



Partial Attribute



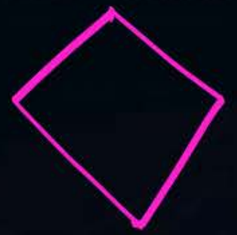
Derived Attribute



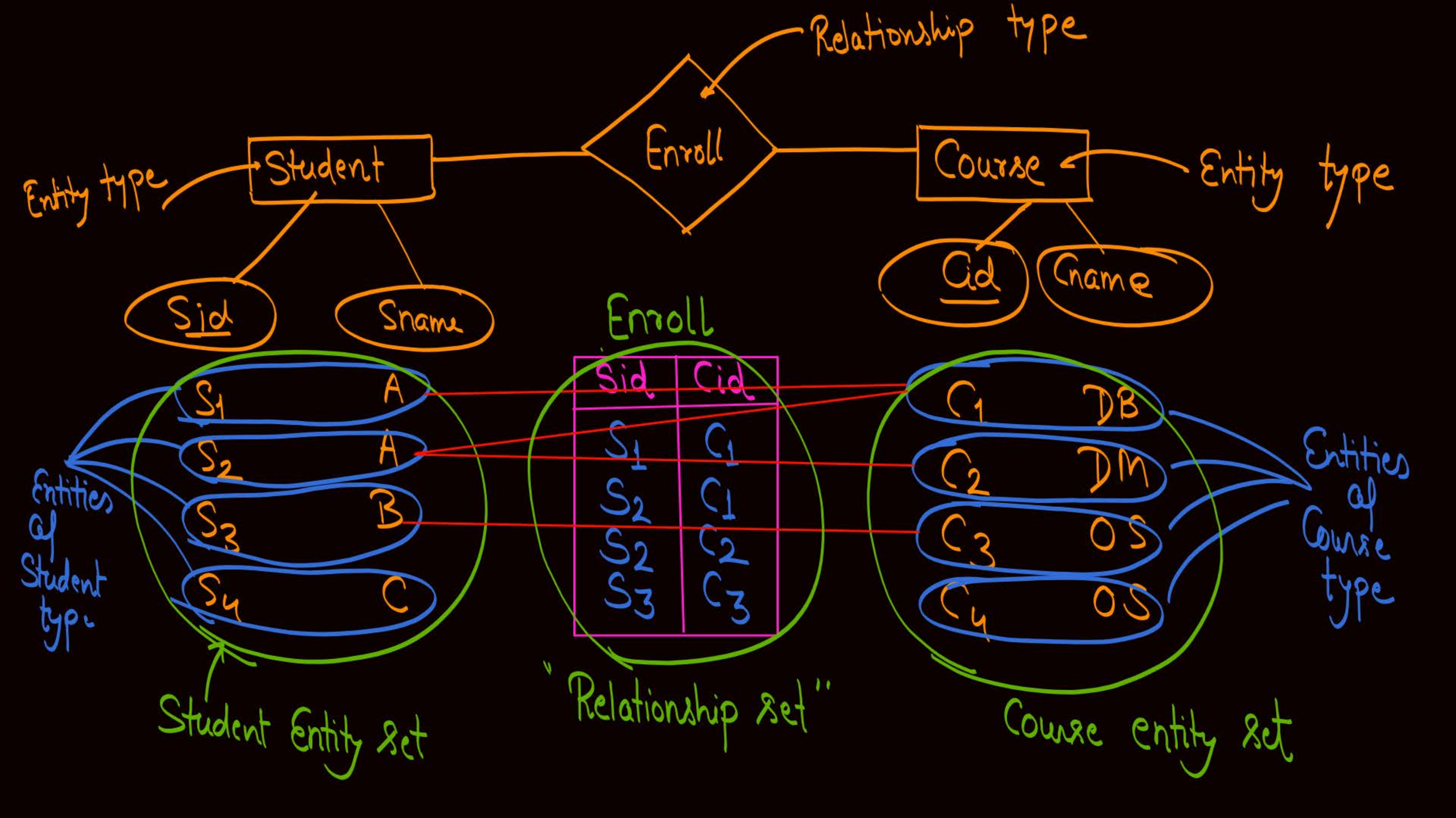


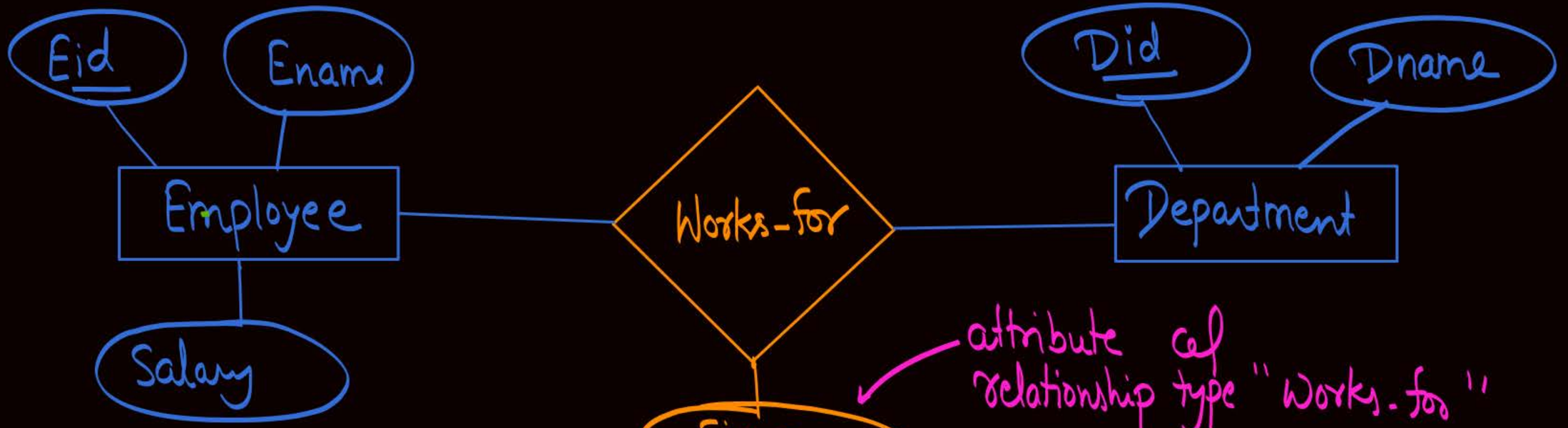
Topic : Relationship in ER model

A Relationship Type represents the association between entity types. For example, 'Enrolled' is a relationship type that exists between entity type Student and Course. In ER diagram, the relationship type is represented by a diamond and connecting the entities with lines.






Relationship type
Enroll
Entity type
Student
Course
Entity type
Sid
Sname
Cid
Cname
Enroll
Sid
Cid
S1
A
S2
A
S3
B
S4
C
C1
DB
C2
DM
C3
OS
C4
OS
S1
C1
S2
C1
S2
C2
S3
C3
Entities of Student type
Entities of Course type
Student Entity set
"Relationship set"
Course entity set



attribute of relationship type "Works-for"

Some times relationship type may have their own attributes

{ Attributes of "relationship type" are called "descriptive attributes"

it will be used to specify that since when the Employee is associated with that department



Topic : Descriptive attribute

- Descriptive attributes are use to provide the information related to relationship set.

Note :- It is not necessary for every relationship type to have their own attributes.



Topic : Degree of a Relationship

- The number of different entity sets participating in a relationship is called the degree of a relationship.

Degree of a relationship = Number of entity sets participating in a relationship



Topic : Types of Relationship

On the basis of degree of a relationship set, a relationship set can be classified into the following types-

i. Unary Relationship: Unary relationship set is a relationship set where only one entity set participates in a relationship set. { It is also known as Recursive Relationship }

ii. Binary Relationship: Binary relationship set is a relationship set where two entity sets participate in a relationship set.

iii. Ternary Relationship: Ternary relationship set is a relationship set where three entity sets participate in a relationship set.

iv. N-ary Relationship: N-ary relationship set is a relationship set where 'N' entity sets participate in a relationship set.

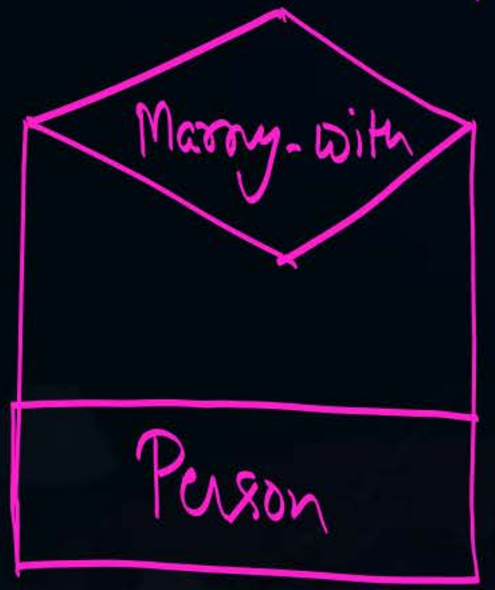


Topic : Unary Relationship



≡

Unary relationship



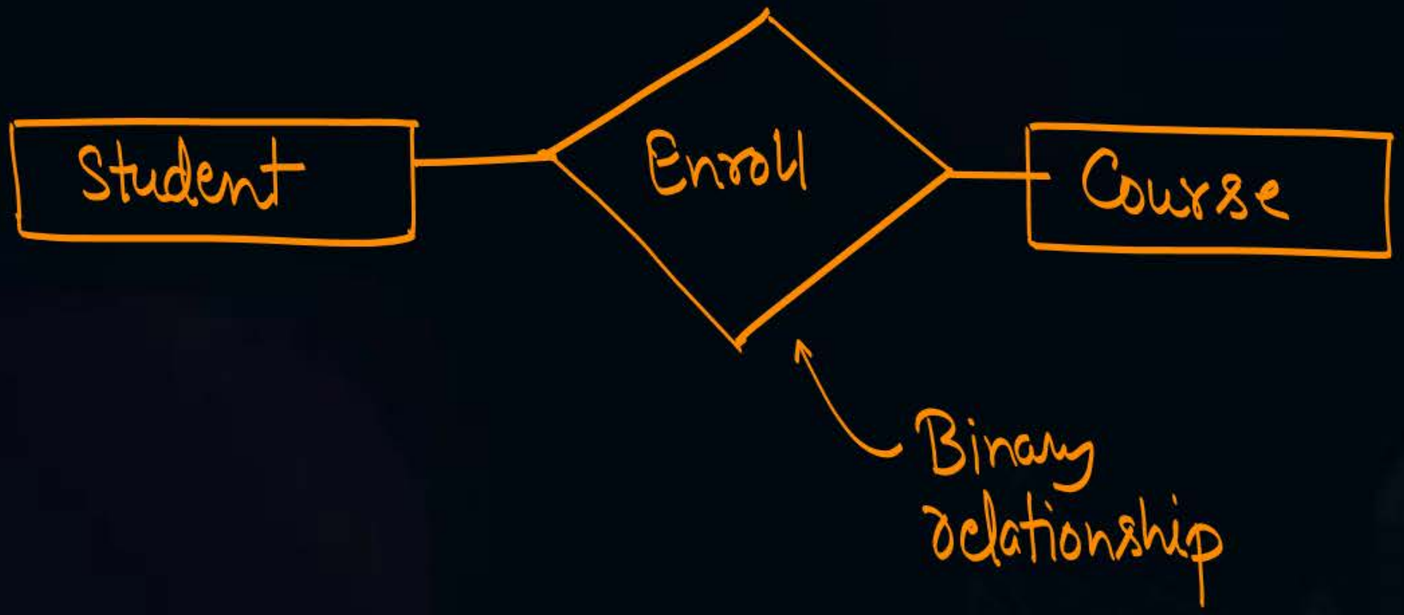
Only one Entity type "Person" is participating in relationship type "Marry-with"

∴ Marry with a Unary Relationship



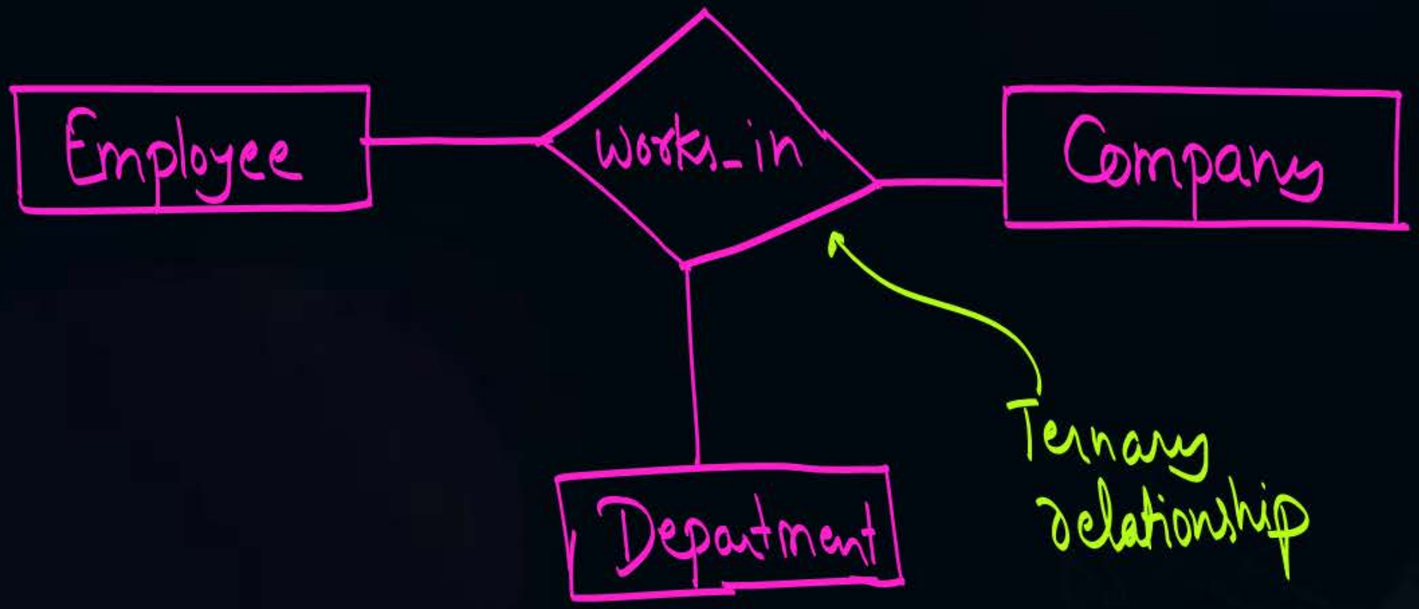
Topic : Binary Relationship

eg:



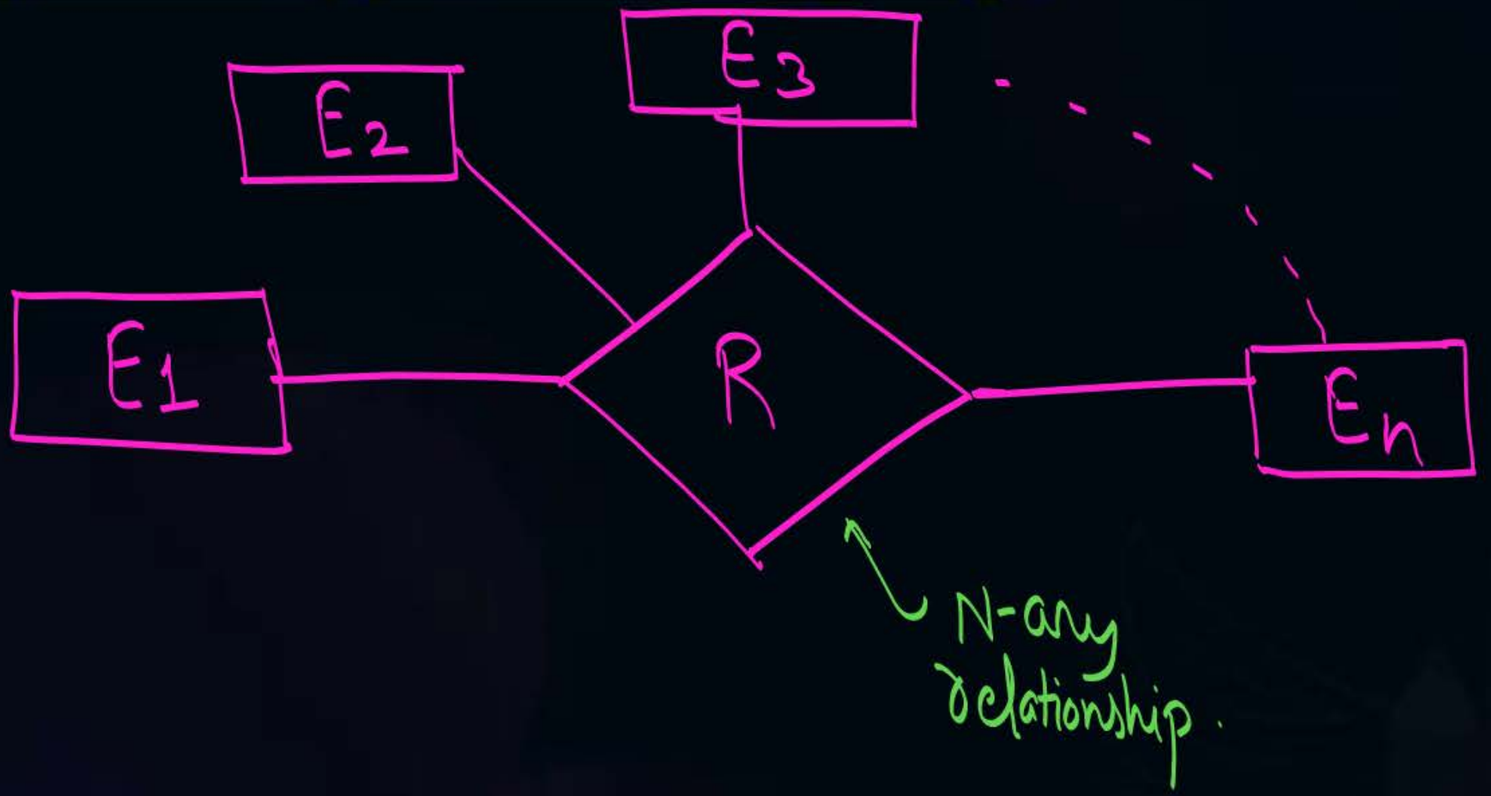


Topic : Ternary Relationship





Topic : N-ary Relationship





Topic : Participation constraints

- There are two types of participation constraints :
- 1. Total participation
- 2. Partial participation



Topic : Total participation

- If every entity of an entity set participate in the relationship set, then that entity set is said to have total participation.
- Total participation of an entity set in a relationship set is denoted by double line



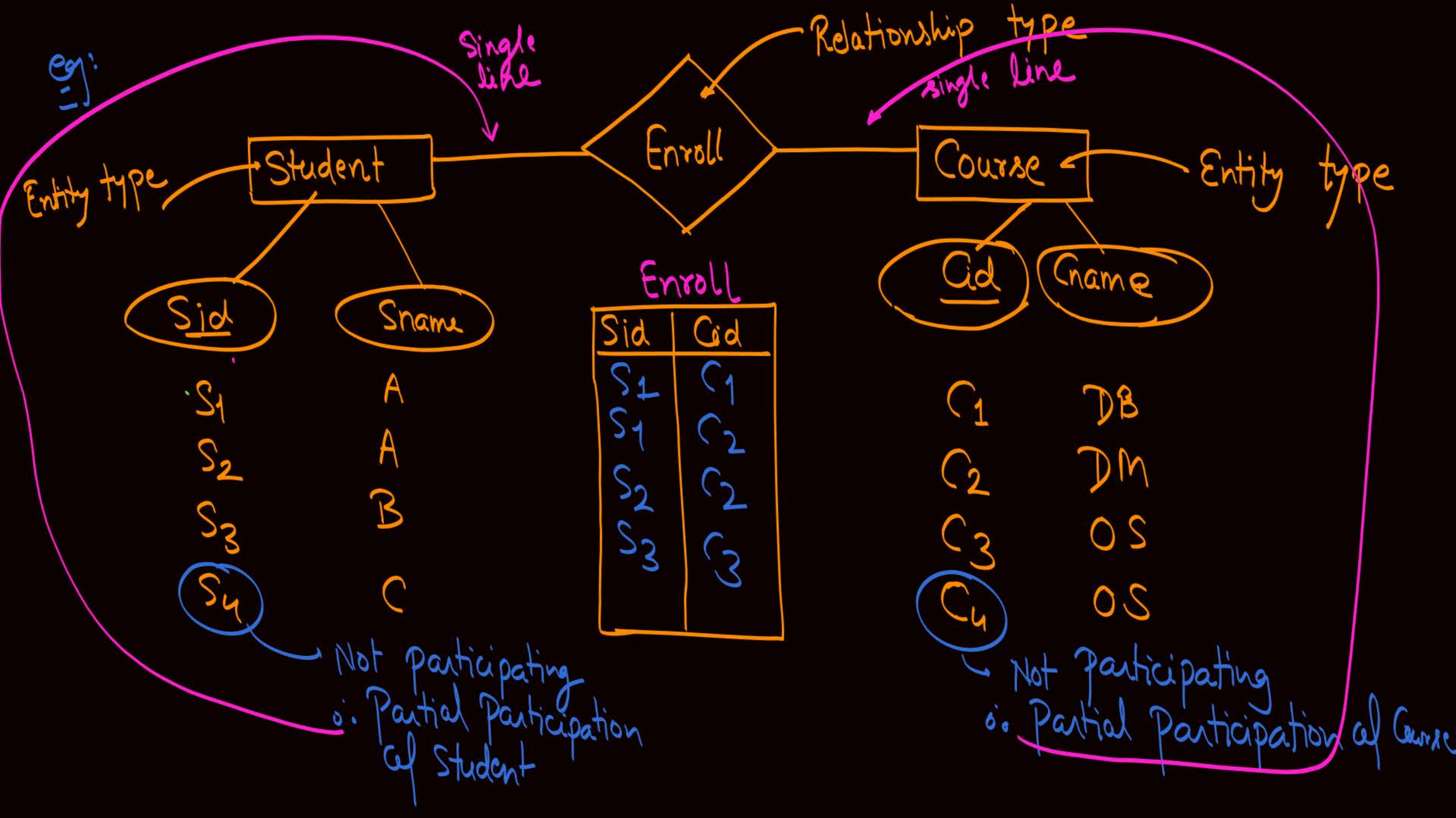
Single line
∴ Partial Participation of E_1 in relationship R

it represent total participation { it represent that Every Entity of Entity Set E_2 Will Participate in relationship R }

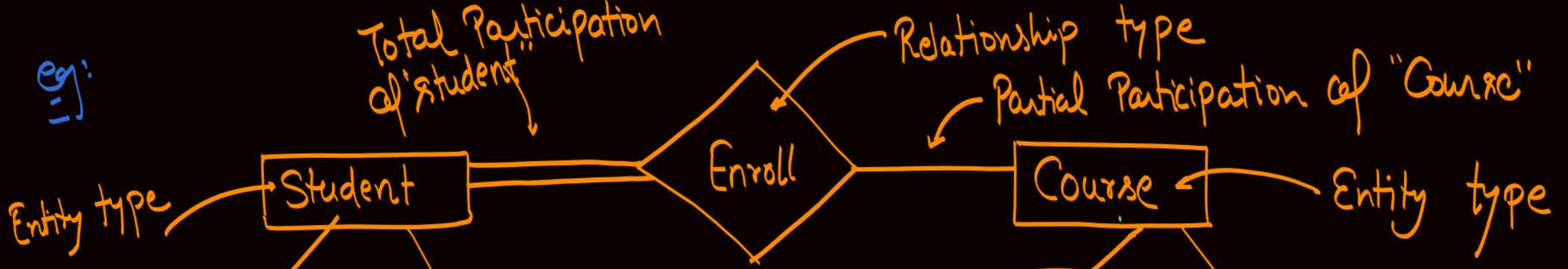


Topic : Partial participation

- If at least one entity of entity set is not participating in the relationship set, then that entity set is said to have partial participation.
- Partial participation of an entity set in a relationship set is denoted by single line.


Eg:
Single line
Relationship type
single line
Entity type
Student
Enroll
Course
Entity type
Sid
Sname
Cid
Cname
Enroll
Sid | Cid
S1 | C1
S1 | C2
S2 | C2
S3 | C3
S1 S2 S3 S4
A A B C
C1 C2 C3 C4
DB DM OS OS
Not participating
i. Partial Participation of Student
Not Participating
i. Partial Participation of Course

eg:-



Sid	Sname
S_1	A
S_2	A
S_3	B
S_4	C

Sid	Cid
S_1	C_1
S_1	C_2
S_2	C_2
S_3	C_3
S_4	C_3

Cid	Cname
C_1	DB
C_2	DM
C_3	OS
C_4	OS

Every Entity of Student Entity set is Participating
∴ Total Participation

C_4 → Not Participating
∴ Partial Participation of Course



2 mins Summary

- Topic: ER model & ER diagram
- Topic: Terminologies in ER diagram

Slide

THANK - YOU